

Lecture 01 — Study Notes

Predicates, Sets, and Proofs

MIT 6.1200J / 18.062J · Spring 2024

§ What is a Proof?

In everyday life the word “proof” means different things: a physics experiment, a legal verdict, a majority opinion, or even “the professor said so.” None of these are mathematical proofs.

Definition — Mathematical Proof

A **mathematical proof** is a **verification** of a **proposition** by a chain of **logical deductions** from a base set of **axioms**.

Three moving parts:

- **proposition** — the statement being proved
- **logical deductions** — each step follows necessarily from the last
- **axioms** — the agreed-upon starting truths

✓ Key Takeaway

A proof is not about convincing; it is about **necessity**. If the axioms are accepted and every deduction is valid, the conclusion **cannot** be false.

§ Propositions and Predicates

Definition — Proposition

A **proposition** is a statement that is unambiguously either **True** or **False**. It has no free variables.

Definition — Predicate

A **predicate** is a proposition whose truth value **depends on one or more variables**. It becomes a proposition once those variables are fixed.

Examples

True $2 + 3 = 5$

False $2 + 3 = 6$

? $\forall n \in \mathbb{N}. n^2 + n + 41$ is prime

The third example is a **predicate** (contains the variable n) wrapped in a universal quantifier — so it is also a proposition. Is it true?

n	$n^2 + n + 41$	Prime?
0	41	✓
1	43	✓
2	47	✓
...	...	...
39	1601	✓
41	43×41	✗ — fails here

⚠ Note

Proof by Example is NOT a proof. Checking finitely many cases — even millions — never proves a universal statement. $n = 41$ kills the proposition above.

Quantifier Notation

Symbol	Meaning
$\forall$	“for all” — universal quantifier
$\exists$	“there exists” — existential quantifier
$\in$	element of a set
$\mathbb{N}$	natural numbers = $\{0, 1, 2, \dots\}$
$\mathbb{Z}$	integers = $\{\dots, -1, 0, 1, \dots\}$
$\mathbb{Z}^+$	positive integers = $\{1, 2, 3, \dots\}$

Devious Propositions

Some propositions remain unresolved for centuries:

Euler’s Sum of Powers Conjecture (1769): $a^4 + b^4 + c^4 = d^4$ has no positive integer solution.

→ **False**. Disproved by Elkies in 1988: $95800^4 + 217519^4 + 414560^4 = 422481^4$.

Goldbach’s Conjecture: Every even integer > 2 is the sum of two primes.

→ **Unknown**. Verified for enormous ranges; no proof exists.

Elliptic curves: $313(x^3 + y^3) = z^3$ has no positive integer solution. Counterexample has > 1000 digits. Elliptic curves underpin modern cryptography — the ability to factor large numbers would compromise most secured systems.

§ Combining Propositions — Boolean Operators

Given propositions A and B we build new ones:

A	$\neg A$
T	F
F	T

A	B	$A \wedge B$	$A \vee B$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Operators Explained

- $A \wedge B$ (AND / conjunction) — both must be true.
- $A \vee B$ (OR / disjunction) — at least one is true. This is **inclusive** or: it is fine if both are true.
- $\neg A$ (NOT / negation) — flips truth value.

⚠ Note

English “or” $\neq$ mathematical $\vee$.

- “Chicken **or** pasta” at a wedding → exclusive or (xor): exactly one.
- “Coffee **or** tea” → neither is also allowed (nand).
- “Cream **or** sugar” → all four combinations valid — this is mathematical $\vee$.

When in doubt, trust the truth table, not the English word.

§ Implication

Definition — Implication $A \Rightarrow B$

Read as “if A then B ”, or “ A implies B ”.

A	B	$A \Rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

Why is “False implies True” = True?

Intuition: “On Wednesdays we wear pink.”

- Wednesday + wearing pink $\rightarrow$ rule satisfied. ✓
- Wednesday + **not** wearing pink $\rightarrow$ rule violated. ✗
- **Not** Wednesday + wearing pink $\rightarrow$ rule not triggered, no violation. ✓
- **Not** Wednesday + **not** wearing pink $\rightarrow$ rule not triggered. ✓

The rule is only **violated** when the hypothesis is true but the conclusion is false.

Four Related Forms

Form	Statement	Equivalent to original?
Original	$A \Rightarrow B$	—
Contrapositive	$\neg B \Rightarrow \neg A$	Yes — always equivalent
Converse	$B \Rightarrow A$	No — independent
Inverse	$\neg A \Rightarrow \neg B$	No (same as converse)

⚠ Note

A common logical error is treating the **converse** as equivalent to the original. “ $x < 3 \Rightarrow x < 4$ ” does NOT reverse to “ $x < 4 \Rightarrow x < 3$ ”.

If and Only If (iff)

A iff B means A and B are either both true or both false. It is defined as $(A \Rightarrow B) \wedge (B \Rightarrow A)$ — implication in both directions.

A	B	$A \text{ iff } B$
T	T	T

T	F	F
F	T	F
F	F	T

§ Sets

Definition — Set

A **set** is an unordered collection of distinct objects (called **elements**).

- No duplicates: $\{1, 2, 2, 3\} = \{1, 2, 3\}$.
- Order irrelevant: $\{1, 2, 3\} = \{3, 1, 2\}$.
- Elements can themselves be sets.

Notation

Symbol	Meaning
$\emptyset$ or $\{\}$	empty set — zero elements
$6 \in A$	6 is an element of A
$x \notin A$	x is not an element of A
$S \subseteq T$	every element of S is also in T (subset)
$A \cap B$	intersection — elements in both A and B
$A \cup B$	union — elements in A or B (or both)
$A \setminus B$	difference — elements of A not in B

Set-Builder Notation

$$\{n \in \mathbb{N} \mid \text{isPrime}(n)\} = \{2, 3, 5, 7, 11, \dots\}$$

Read: “the set of all n in $\mathbb{N}$ such that n is prime.” A colon ($:$) may appear instead of $\mid$ — both are standard.

Sets vs. Ordered Tuples

Set $A = \{6, 1, 2, 0\}$

- Unordered
- No duplicates
- Supports $\cup, \cap, \setminus$

Tuple $X = (6, 1, 2, 0)$

- Ordered
- Duplicates allowed: $(1, 1, 2)$
- No union/intersection

§ Axioms

Definition — Axiom

An **axiom** is a proposition assumed to be true without proof. It is the foundation on which all theorems in a system are built.

Every proof rests on axioms. The rule is not “make no assumptions” but “**state** your assumptions explicitly.”

Different Axioms → Different Geometries

The **parallel postulate** has three mutually contradictory versions:

Axiom	Statement	Geometry
2	Exactly one parallel line through p	Euclidean
3	No parallel line through p	Spherical / Elliptic
4	Infinitely many parallel lines through p	Hyperbolic

All three are internally consistent. None is “more correct” — they describe different geometric worlds.

Desirable Properties of an Axiom System

Definition — Consistency

A set of axioms is **consistent** if no proposition can be **both** proved and disproved within the system.

Definition — Completeness

A set of axioms is **complete** if every proposition can be **either** proved or disproved within the system.

Both properties are obviously desirable. Russell and Whitehead spent decades trying to build a complete and consistent foundation for arithmetic in their *Principia Mathematica*.

Gödel’s Incompleteness Theorem

Theorem (Gödel, 1930s)

No set of axioms is both **complete** and **consistent**.

Consequences:

- If you insist on consistency (you must), then there will always be true statements that are **unprovable** within your system.
- Goldbach's Conjecture might be one such statement — true, yet forever beyond proof.
- This result **shocked** the mathematical community and effectively ended the Russell–Whitehead programme.

✓ Key Takeaway

Consistency is non-negotiable: a system where P and $\neg P$ are both provable is useless — you could prove anything. Completeness is the price we pay.

§ Quick Reference — Symbol Glossary

Symbol	Name	Meaning
$\wedge$	AND	both true
$\vee$	OR	at least one true (inclusive)
$\neg$	NOT	negation
$\Rightarrow$	implies	if ... then ...
iff	iff	if and only if
$\forall$	for all	universal quantifier
$\exists$	there exists	existential quantifier
$\in$	element of	$x \in S$ means x belongs to set S
$\notin$	not element of	—
$\subseteq$	subset	$S \subseteq T$: every element of S is in T
$\cap$	intersection	elements common to both sets
$\cup$	union	all elements from either set
$\setminus$	set difference	$A \setminus B$: elements in A but not B
$\emptyset$	empty set	set with no elements
$\mathbb{N}$	naturals	$\{0, 1, 2, \dots\}$
$\mathbb{Z}$	integers	$\{\dots, -1, 0, 1, \dots\}$
$\mathbb{Z}^+$	positive integers	$\{1, 2, 3, \dots\}$

§ Common Mistakes to Avoid

- **Proof by example** — checking cases, even millions, does not prove a universal statement.

- **Confusing implication with causation** — $A \Rightarrow B$ says nothing about A causing B , only about their truth values.
- **Treating the converse as equivalent** — $A \Rightarrow B$ and $B \Rightarrow A$ are independent. Only the **contrapositive** $\neg B \Rightarrow \neg A$ is equivalent to $A \Rightarrow B$.
- **Using inclusive or when you mean exclusive or** — mathematical $\vee$ allows both; everyday English “or” often doesn’t.
- **Assuming completeness** — Gödel guarantees there are truths that no axiomatic system can prove. Some open conjectures may fall here.